

**Comparative Analysis of the Effects of Static Magnetic Force on *Tetraselmis chuii*'s
Biomass Production and Nitrogen Removal Efficiency**

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STEM English 12 Period 1

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March 3rd, 2024

Role of Student and Mentor

The mentor recommended and demonstrated how to use the spectrophotometer. The mentor recommended purchasing the algae culture from Algae Research Supply. Student designed and created the magnet field apparatus for 300 Gauss, 600 Gauss, and 900 Gauss. The student developed and implemented each step of the procedure.

Introduction

Harmful algal blooms or HABs have increasingly become an issue for coastal nations, as they destroy the local marine ecosystem. This leads to enormous damage to the local fishing industries, and according to the National Centers for Coastal Ocean Science, the average annual cost of HABs in the United States is around \$10-100 million. The prevalence of HABs have only grown in recent years, and one of the reasons for the HABs growth has been the increase in nutrients in effluents and agricultural runoffs. The nitrogen and phosphorus in this wastewater enter the ocean and help stimulate HABs growth (Anderson et al., 2012). Therefore, to reduce the level of HABs growth, we must address the wastewater entering the ocean.

Review of Literature

One of the most popular water treatment processes currently is the conventional activated sludge, or CAS, process. It uses pre-treated wastewater with microorganisms that will remove nutrients from the wastewater. The reduced and reactive nitrogen in the wastewater is then converted into N_2 gas through the denitrification process which leads to N_2O greenhouse gas emissions, representing 26% of total water chain (van der Hoek et al., 2018). as well as a loss of valuable nitrogen which could've been reused in agriculture. This greenhouse gas emission will then contribute to warming the Earth's climate and in turn, contribute to HABs growth. Duijff et al. (2018) attempted to recover nitrogen from an existing wastewater treatment plant CAS

process. Utilizing struvite precipitation, the treatment of digester rejects water by air stripping, vacuum membrane filtration, hydrophobic membrane filtration, and treatment of air from thermal sludge drying. The treatment yielded a low of 1.1% nitrogen recovery and a high of 75% nitrogen recovery, however, the effect of sustainability was low. The article concluded that radical improvement in nitrogen recovery technology was needed before it could become a viable addition to the CAS process.

An alternative to the CAS process is the microalgal biofilm process. Boelee et al. (2012) investigated the microalgal biofilm treatment process in three scenarios: post-treatment, second-stage treatment, and an isolated microalgae/heterotrophic system. In each of these scenarios, the microalgal biofilm would remove nitrogen and phosphorus from the wastewater. Results of the scenario showed that it wasn't possible to remove both N and P in the wastewater to the target values of 2.2 mg N/L and 0.15 mg P/L. The study further found that microalgae growing on treated wastewater was phosphorus-limited, while untreated wastewater was nitrogen-limited. The area requirement for each scenario also differed, with scenario one having an area requirement of 0.32 m²/PE, scenario 2 with 2.1 m²/PE, and scenario 3 with 0.76 m²/PE. Comparatively, scenarios 2 and 3 require a lot more area to treat an equivalent amount of people than a typical wastewater treatment plant at 0.2 to 0.4 m²/PE. Finally, the study found that this microalgal biofilm treatment process resulted in a large quantity of biomass, which should be investigated for its applications.

As identified in the previous study, microalgae can be a wastewater treatment solution, however, it needs to become more efficient before it's commercially viable. To improve the efficiency of microalgae in wastewater treatment, four microalgae species were cultivated in unsterilized effluent of a livestock anaerobic digestion unit, to identify the species best suited for

biomass productivity and nutrient recovery. The four microalgae species used were *Chlorella vulgaris*, *Chlorella sorokiniana*, *Desmodesmus communis*, and *Stichococcus sp.* contained in a 500ml Erlenmeyer flask with BG-11 medium containing nutrients (Psachoulia et al., 2022). The study showed that anaerobic digestion effluent could serve as a growth medium for microalgae cultivation. It was found that *Chlorella vulgaris* resulted in the highest biomass production when cultivated in BG-11 and that ammonium (N-NH₄) was the preferred nitrogen form for all investigated species. *Stichococcus sp.* was found to assimilate ammonium nitrogen most rapidly, while *Desmodesmus communis* recorded the highest phosphorus removal rate (Psachoulia et al., 2022).

Beyond finding the most efficient algal species for nitrogen removal and biomass usefulness, there's still the need to improve the efficiency of the algal biofilm process in order for it to be economically viable and achieve mass adoption. In order to help improve this efficiency, one study investigated the effects of different wavelengths of light on algae biomass growth. The concept behind using different wavelengths is the effect it has on photosynthesis. Plants grow through a photosynthesis process, where pigments, primarily chlorophyll a and b in green algae, capture light with a radiation range of around 400 to 700 nm, carbon dioxide, and water to grow (Psachoulia et al., 2021). The chlorophyll bands a and b absorb mainly blue (430 - 475 nm) and red (630 - 680 nm) light. It was found that exposure of *Chlorella vulgaris* cultures to blue light led to significantly increased cell size, while exposure to red light resulted in a maximum number of smaller cells (Kim et al., 2014).

Another method to improve algae growth can be to treat the algae culture with static magnetic fields (SMF). The theory is that the magnetic field (MF) can help promote the release of oxygen from algae which increases the photosynthesis rate, improving metabolism and growth

rate (Luo et al., 2020). Luo et al. (2020) investigated the effect of 0 Gauss (G), 200 G, 400 G, 800 G, to 1500 G on the growth of *Chlorella vulgaris* over 16 days. It was found that *Chlorella vulgaris* treated with 800 G magnet grew at the fastest rate and produced the greatest amount of biomass. However, it also found that *Chlorella vulgaris* samples treated with excessive amounts of SMF strengths for an excess amount of time inhibited the growth of *Chlorella vulgaris* (Luo et al., 2020). Deamici et al. (2016) investigated the same effect on *Spirulina sp.* using 30 mT, 300 G, and 60 mT, 600 G. This study ran their experiment over 15 days, and came to a similar conclusion finding that as MF increases, so does the concentration of *Spirulina sp.* biomass.

In order for these microalgal wastewater treatments to be sustainable, there has to be a use for their biomass. Historically, *Chlorella* and *Spirulina* algae species biomass have been grown for direct consumption and as a food supplement. However, recently, the *Tetraselmis* algae species have been investigated for their nutritional contents and as a source of food and animal feed. This algal species's biomass displays a similar composition to soybean and possesses high levels of carbohydrates compared to other algae species (Pereira et al., 2019). *Tetraselmis chuii* specifically has been used to replace wheat flour in protein-enriched bread, and in buckwheat, rice flour, and potato starch formulations for gluten-free bread. With EU authorization for commercialization as a novel food and food supplement, *T. chuii* is becoming a promising ingredient for the food industry (Patterson et al., 2023). Beyond its nutritional content, it's also efficient in nutrient removal in wastewater. *T. sp.* has a higher growth rate and nutrient uptake, when compared to *Dunaliella sp.* and *Nannochloropsis sp.*, removing around 98% of total nitrogen (Sacristán de Alva et al., 2018).

Objective

There has been little research done with *T. chuii* optimized for nitrogen and phosphorus removal as well as optimized biomass production. Furthermore, there has been little to no research on the effect of SMF on *T. chuii* growth. This research aims to investigate the effect SMF has on the *T. chuii* microalgae species. Specifically, this study aims to determine SMF effect on the nitrogen and phosphorus removal rate of *T. chuii* and its biomass production.

The SMF strengths used in this study will be 300 G, 600 G, and 900 G. These ranges was chosen because its been shown that *C. vulgaris* growth rate was highest at 800 G, however above that around 1400 G, growth of *C. vulgaris* was inhibited (Luo et al., 2020). As for time of application, it was found that *Spirulina sp.* applied with 600 G SMF for 1 hd^{-1} had a greater biomass concentration than both the control and 24 hd^{-1} (Deamici et al., 2016). Following these experimental conditions, the time of application per day for each *T. chuii* algae culture will also be stimulated by SMF for either 1 hd^{-1} or 24 hd^{-1} . Using these experimental conditions, this study aims to answer the following research questions:

Problem Statement

- How do the different SMF strengths affect the growth, biomass production, and nitrogen removal of *Tetraselmis chuii*?
- How does the time of application of SMF affect the growth, biomass production, and nitrogen removal of *Tetraselmis chuii*?
- Does prolonged SMF treatment of *Tetraselmis chuii* have a negative impact on the growth, biomass production, and nitrogen removal of *Tetraselmis chuii*?

Hypothesis

1. Although there is still much debate over the physiology of the effects of static magnetic field (SMF) on microorganisms and plants, SMF has been known to affect

microorganisms' photosynthesis process and metabolism by promoting the release of oxygen and changing its enzyme activity. When tested with other types of algae, SMF has been seen to improve the algae's biomass concentration and nutrient uptake. Thus, SMF should also, theoretically, promote the growth, biomass production, and nitrogen removal efficiency of *Tetraselmis chuii*.

2. 24 hd^{-1} of SMF treatment should have a greater effect on the growth, biomass production, and nitrogen removal of *Tetraselmis chuii* compared to its counterpart 1 hd^{-1} , since there's a greater exposure period.
3. With an increased growth rate, the algae should take up more nutrients exhausting the supply of nutrients faster. This should then lead the growth, biomass production, and nitrogen removal of *Tetraselmis chuii* to be negatively impacted near the end of the experimental timeline.

Methodology

1. Preparation

1.1 Culture and Media Preparation

The culture's media was prepared according to the instructions of Algae Research Supply Algae Culturing Manual. 1 bag of culture salt was added to 500 mL of purified water, which according to the manual would have the correct balance of pH and nutrients for optimal algae growth. The culture media was then added, along with 250 mL of the algae culture, into a larger 1 L container. From there, the container was mixed until homogenous and 80 mL was poured into each of the seven 100 mL containers.

1.2 Static Magnetic Field Apparatus Set Up

To achieve the magnetic field strength of 300, 600, and 900 gauss, ceramic magnets and a teslameter was used. For the 300 gauss strength, ceramic magnets were separated until each teslameter read 30 mT equivalent to 300 gauss, with magnets being added as required. The distance between the ceramic magnets was then measured and then used as the outer diameter of a cylindrical frame which would separate the magnets. This was repeated for each of the other magnetic field strengths. Once the frames were created, -- the frames were created using the computer-aided design software, Corel, and a laser cutter -- the ceramic magnets were then taped to the frames, holding them in place.

2. Experiment

2.1 Measurement/Data

For each day of the experiment, 2 mL of each sample was extracted and measured using a spectrophotometer. The wavelength used was 600 nm as it would best measure algae growth while avoiding absorbent material, according to the mentor. The spectrophotometer was first calibrated using water as a blank and adjusted to 100 transmittance. Then the blank was removed and replaced with the 2 mL culture sample and the spectrophotometer was switched over to absorbance. The absorbance value would then be recorded.

Nitrogen levels were measured using nitrate/nitrite test strips. To measure, the test strips were dipped into the sample for 3 seconds and left for 30 seconds to change in color. The color would then be measured using the color scale on the nitrate/nitrite test strip container, listing the concentration of nitrate or nitrite in ppm correlated to the color. These values were then recorded.

3. Results/Discussion

Each sample's concentration started at 0.2 absorption and 500 ppm nitrate level. As seen in Figure 1, each culture's concentration would meet or exceed the concentration of the control,

with the exception of 300 gauss 1hd^{-1} . Both of the 900 Gauss cultures exhibited slightly greater peak absorption at 0.24 compared to the control's peak at 0.23, however, this difference is not statistically significant, with the difference being under 5%.

The SMF strength that did show statistically

significant growth over the control was 600 gauss for 24hd^{-1} , resulting in the greatest absorption

of 0.25 compared to 0.23 of the control which

represents an 8.7% higher growth than the control.

Therefore, 600 gauss in isolation would validate

Hypothesis 1, however, including the other culture samples, Hypothesis 1 cannot be validated.

Furthermore, for each culture, the culture with a

longer exposure time to SMF had a greater peak biomass concentration than its counterpart.

However, despite each 24hd^{-1} having higher growth, it was not significantly higher, with less than a statistically significant 5%, therefore, Hypothesis 2 can't be comprehensively validated,

that a longer exposure time to SMF does increase the

growth rate of algae. As for the algae entering the decline phase, all samples entered the decline phase around the same time.

Throughout the experiment, the nitrate/nitrite test strips did not detect a decrease in nitrogen concentration. This is likely because the algae did

Figure 1: T. Chuii Biomass Concentration

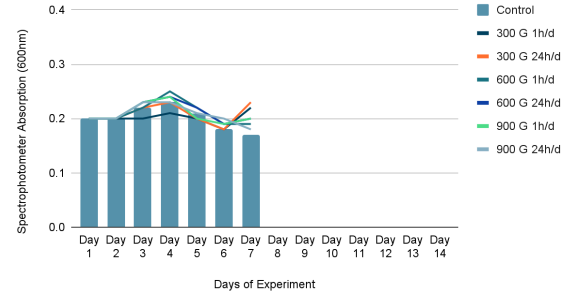


Figure 2: Nitrogen Concentration

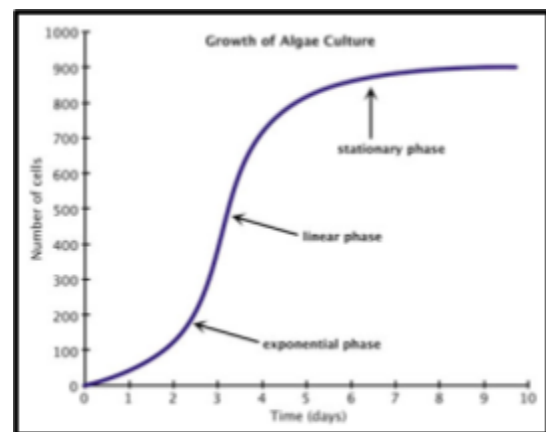
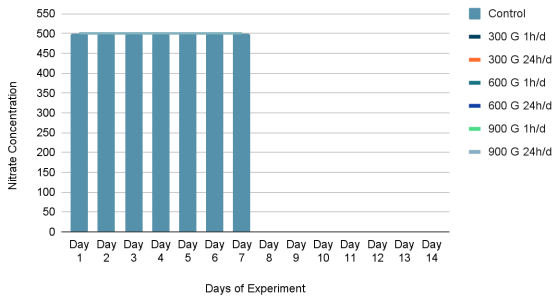


Figure 3: Typical Algae Growth Phases

not grow significantly to remove a significant amount of nitrogen.

However, the issue with all of this data set is that it doesn't meet the expected growth rate, period, and phases of the algae. The growth period of the algae should last longer than three days -- expected growth period is around 10 days as seen in Figure 3 -- before entering stagnation and decline, and the growth rate should be far greater than the high of 20% increase in 600 gauss 24 hd⁻¹.

The unexpected data sets are likely due to the culture failing to grow properly. After discussing with the mentor, it could be due to the water temperature being too low during certain periods of the day, falling well below 25 degrees Celsius which is optimal for *Tetraselmis* growth, it could also be due to insufficient lighting due to weather conditions as the main source of lighting for this culture was natural sunlight. *Tetraselmis* along with other types of algae such as *Spirulina* and *Chlorella* requires ample sunlight to grow, and the weather conditions during the time of experiment included cloudy, rainy, snowy, and thunderstorms.

If this experiment were to be redone, the algae cultures should be in a specific and consistently warm environment of around 20 degrees Celsius, under a consistent white fluorescent light for 12 hours a day to emulate the day/night cycle of the sun, keeping all other variables the same.

4. Conclusion

Although 600 Gauss for 24 hd⁻¹ showed a statistically significant increase in growth rate compared to the control, at 8.7% higher than the control, the rest fell below 5% higher, therefore Hypothesis 1 which states that algae under the effect of SMF will have a greater growth rate, nitrogen extraction, and biomass concentration cannot be validated.

Hypothesis 2, which states that algae under the effect of SMF for a longer period of time will exhibit greater growth rate, biomass concentration, and nitrogen removal rate, also cannot be validated. Each of the 24 hd^{-1} did show a higher growth rate, however, it was higher by less than 5%, around 2 to 3%, and therefore is not statistically significant.

Hypothesis 3, which states that extended exposure to SMF will result in algae entering the decline phase faster is also not validated, since each culture entered the decline phase at the same time.

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